Dell PowerFlex 3.6.3 Release Notes

PowerFlex 3.6.3

Rev. 01
March 2024
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Revision history

Revision history for PowerFlex3.6.3 release notes

Table 1. Revisions

Revision	Date	Description
01	February 29, 2024	Release of PowerFlex 3.6.3

Product description

PowerFlex is a software-defined storage platform designed to significantly reduce operational and infrastructure complexity. It empowers organizations to move faster by delivering flexibility, elasticity, and simplicity with predictable performance and resiliency at scale.

The PowerFlex family provides a foundation that combines compute as well as high performance storage resources in a managed unified fabric. PowerFlex comes in fully managed flexible deployment options for rack or appliance that enable Server SAN, HCI, and storage only architectures. PowerFlex can now be deployed in the public cloud. A Ready Node option is also available. PowerFlex is ideal for high performance applications and databases, building an open private cloud, or consolidating resources in heterogeneous environments.

New and changed features

Learn about new and changed features in this version of PowerFlex.

New features for PowerFlex

Learn about the new features introduced in PowerFlex 3.6.3 software.

- Storage Data Client self-compiles for existing operating systems—In PowerFlex 3.6.3, an SDC self-compile capability is enabled that enables creation of a driver kernel to match the operating system. This process allows customers who want to update the operating system and compile the SDC driver to match the kernel version of the operating system.

Important

The following aspects of PowerFlex 3.6.3 must be understood before deploying or upgrading your system:

1. CloudIQ integration
   - Discovering a PowerFlex system in CloudIQ requires the system to be registered with Secure remote services. In the PowerFlex Installer, **Maintain** tab, ensure that the PowerFlex system is registered with Secure remote services and that **Send data to CloudIQ** is enabled.
   - After installing a *new* PowerFlex system with PowerFlex version 3.6, it takes 24 hours until data is available in CloudIQ. Systems *upgrading* to PowerFlex 3.6 will appear in CloudIQ within an hour of enabling the **Send data to CloudIQ** option.
   - PowerFlex appliance and rack systems need PowerFlex Manager version "H" in order to be discovered in CloudIQ.
2. Replication enhancements
   - Bi-directional replication between two sites only is supported.
   - See the PowerFlex Sizer for sizing guidelines for correct replication system specifications, including WAN bandwidth sizing. When using the PowerFlex Sizer, consider existing workload and existing hardware, and the possible need to add additional nodes to enable replication without impacting the current workload.
   - Replication capability has higher I/O overhead, as explained in the v3.6 *Getting to Know PowerFlex Guide* (Replication section) and the white paper Dell EMC PowerFlex: Introduction to Replication. See these recommendations in order to size your system correctly, and ensure that you have sufficient existing storage capabilities to enable this ability.
3. Network resiliency improvements
   - Proactive network path error detection requires SDC to be at v3.6 or later.
4. Upgrade Paths

- Upgrade from PowerFlex version 3.0.x and 3.5.x to version 3.6 are supported.
- Older PowerFlex/ VxFlex operating system versions must be upgraded to version 3.0.x prior to upgrading to version 3.6.

5. vSphere plug-in End-of-Support-Life
   - The PowerFlex vSphere plug-in continues to be supported with vSphere versions 6.5 and 6.7, however it reaches end-of-support-life when vSphere 6.5 and 6.7 reach end-of-support by VMware. For details on the last release of the vSphere plug-in due to end of general support for ESXi 6.5 and 6.7 by VMware, see: Announcing End of General Support for vSphere 6.5.x and vSphere 6.7.x (89305). Also note the new issue in the *Known Issues* section that is related to vSphere Plug-in.
   - The PowerFlex vSphere plug-in does not support vSphere 7 hyperconverged (HCI) configurations.

Changed features

Learn about the enhanced features for this version of PowerFlex.

Updated operating system support matrix

The following operating systems are added to the support matrix in this release:

- Red Hat Enterprise Linux 8.9
- Red Hat Enterprise Linux 9.3

The operating systems support software only customers for the following components:

- MDM
- SDS
- SDR
- SDC
- LIA

Further to SDC only, the following operating systems support are being added:

- vSphere 8.0U2

For more operating system-related information and supported components, see the PowerFlex simple support matrix: https://elabnavigator.dell.com/eln/modernHomeSSM

Oracle JRE and OpenJDK support

- OpenJDK 11.0.12.07 (Linux-based servers only)
- Presentation server—The Java version must be upgraded to Java 11 before upgrading the presentation server.

NOTE: This patch version of PowerFlex Manager does not support Java version 1.8. Dell Technologies recommends upgrading to Java version 11 for PowerFlex Gateway or Installation Manager.

Support deprecation

Learn about the features that are no longer supported in PowerFlex 3.6.3.

Background scanner events and alerts

Some events and alerts for the background scanner feature were deprecated, and replaced with different ones. See the *PowerFlex v3.5.1.2 Release Notes*, "Documentation" section, for details.

DAS Cache

DAS Cache is no longer supported for PowerFlex running on any operating systems.

System Analysis Tool

The System Analysis Tool is no longer included in PowerFlex.

AMS

Systems managed by AMS system cannot be upgraded to PowerFlex v3.5 or later. Prior to performing an upgrade, contact PowerFlex Product Management for possible alternatives.

NOTE: AMS artifacts are removed from the support site. For guidance, contact PowerFlex Product Management.

VxFlex OS UI client

The UI from versions earlier than v3.5 of this product is no longer supported and cannot be upgraded. From v3.5 and later, a new UI must be deployed or upgraded instead.

Deprecated operating systems

For the most current information, see the *Dell Technologies Simple Support Matrix* at https://elabnavigator.dell.com/eln/modernHomeSSM.

Fixed issues

The following table lists the issues that are fixed in PowerFlex3.6.3.

NOTE: If an issue was reported by customers, the customers' Service Request numbers appear in the "Issue number & SR number" column, and correlate between customer-reported issues and the PowerFlex issue number.

Table 2. Fixed issues

Issue number & SR number	Problem summary
SCI-136684	Improper Input Validation vulnerability in Apache Tomcat. ApacheTomcat from 11.0.0-M1 through 11.0.0-M10, from 10.1.0-M1 through 10.1.15, from 9.0.0-M1 through 9.0.82 and from 8.5.0 through 8.5.95 did not parse HTTP trailer headers correctly.
SCI-136293 SR# 180172024	SDS reporting fixed read errors during and after a persistent checksum rebuild.
SCI-135597	Presentation service is using Jetty 9.4.49.v20220914 which is exposed to CVE-2023-44487 and hence, the Jetty is upgraded.
SCI-133646 SR# 177556520	After performing the CRS procedure for 132 combs, 48 combs remained in an unhealthy "Normal" state. In this state, the comb was Normal, but the secondary device was in a failure state, which is the cause of the CRS.
SCI-133618 SR# 1067101	Vulnerability detected during SQL injection on the presentation server. Fixed security issue - SQL injection on the presentation server.
SCI-78209	Upgrade Tomcat to the latest 9.x version owing to EOL of tomcat 8.5.x and security issues.
SCI-78046 SR# 175576474	MDMs failed and switched over after SDSs decoupled.

Table 2. Fixed issues (continued)

Issue number & SR number	Problem summary
SCI-77521	The command, "--set_mdm_credentials" command through `FOSGWTool.sh` over Red Hat Enterprise Linux 8 and 7 fails with OrcaleJRE while lockbox exists.
SCI-76505	After PowerFlex is upgraded to 3.6.700 from 3.0.1400, one SDS fails to exit IMM. The command, 'pkill sds' helps exit IMM.
SCI-67977	PowerFlex Gateway 3.6.0.4 fails to collect logs from SDC running on ESXi 7.x.

Software packages

Learn about the contents of the PowerFlex 3.6.3 software packages.

Download location

The PowerFlex 3.6.3 complete software package, containing all the packages that are listed below, can be downloaded from the following location:

https://www.dell.com/support/home/en-us/product-support/product/scaleio/overview.

Table 3. Software packages

Package Name	Operating System	GW	MDM	SDS	SDR	SDC	LIA	Xcache	Content
Red Hat Enterprise Linux_OEL7	Red Hat Enterprise Linux / CentOS / Oracle Linux	X	X	X	X	X	X	X	Core components for Red Hat Enterprise Linux, CentOS, and Oracle Linux 7.x. Oracle Linux 7.8 and higher includes KVM.
Red Hat Enterprise Linux 8_OEL8	Red Hat Enterprise Linux / CentOS / Oracle Linux	X	X	X	X	X	X	X	Core components for Red Hat Enterprise Linux, CentOS, and Oracle Linux 8.x. Oracle Linux includes KVM.
Red Hat Enterprise Linux_OEL9	Red Hat Enterprise Linux / CentOS / Oracle Linux		X	X	X	X	X		Core components for Red Hat Enterprise Linux, CentOS, and Oracle Linux 9.x.
CENTOS 7.9									
CentOS Stream 8		X	X	X	X	X	X	X	Core components for CentOS Stream 8
CentOS Stream 9			X	X	X	X	X		Core components for CentOS Stream 9
SLES12.5	SUSE Linux	X	X	X	X	X	X	X	Core components for SLES 12 SP5
SLES15.3	SUSE Linux	X	X	X	X	X	X	X	Core components for SLES 15 SP3
SLES15.4	SUSE Linux		X	X	X	X	X		Core components for SLES 15 SP4
SLES15.5	SUSE Linux		X	X	X	X	X		Core components for SLES 15 SP5

Table 3. Software packages (continued)

Package Name	Operating System	GW	MDM	SDS	SDR	SDC	LIA	Xcache	Content
Ubuntu18.04	Ubuntu	X	X	X	X	X	X	X	Core components for Ubuntu 18.04
Ubuntu20.04	Ubuntu	X	X	X	X	X	X	X	Core components for Ubuntu 20.04
Ubuntu22.04	Ubuntu		X	X	X	X	X		Core components for Ubuntu 22.04.x LTS
XEN 7.1	XenServer / Citrix Hypervisor		X	X	X	X	X		Core components for Citrix XenServer 7.1.2 LTSR
XEN 8.0			X	X	X	X			Core components for Citrix Hypervisor 8.1 and 8.2
Complete_VMware_SW	vSphere	X	X	X	X	X			Contains all PowerFlex, VMware-related software artifacts (OVA file, drv_cfg file, SVM updating, SDC package, vSphere plug-in, and VASA provider) **NOTE:** PowerFlex vSphere plug-in is not compatible with ESXi 7.0 and later.
Complete_Windows_SW	Windows Server	X				X	X		Contains all Microsoft Windows (2012-2019) server-related software artifacts, including the gateway and CSV deployment templates
Red Hat Enterprise Linux CoreOS	CoreOS					X			SDC component for Red Hat Enterprise Linux CoreOS
CoreOS						X			SDC component for CoreOS
AIX7.2	AIX					X			SDC component for IBM AIX 7.2
Presentation_Server									Contains PowerFlex WebUI server (Linux and Ubuntu)
Gateway_for_Linux		X							PowerFlex Gateway, which includes the following: Installer, Gateway (for REST API, Alerts, SNMP/ Secure Remote Services) for Linux/ Ubuntu (can be used to deploy

Table 3. Software packages (continued)

Package Name	Operating System	GW	MDM	SDS	SDR	SDC	LIA	Xcache	Content
									PowerFlex on Linux-based operating systems only). CSV deployment templates are also included.
GPG-RPM-KEY									Contains RPM key (RSA for Linux, DSA for Xen) to authenticate the RPM packages using manual install/upgrade procedure.

NOTE:

- See the latest PowerFlex simple support matrix at https://elabnavigator.dell.com/eln/modernHomeSSM
- PowerFlex version 3.5 and later provides a Web UI. User interface packages are no longer available.

Install and upgrade

PowerFlex 3.6.3 can be installed on a clean system or used as an upgrade. Learn important information about installing or upgrading to this version.

Installation

For complete instructions for installing PowerFlex, see the *Deploy PowerFlex Guide*.

Licensing

PowerFlex use requires a license. Call your Dell Technologies account representative to purchase a license for PowerFlex.

Upgrade

The following table lists the versions from which the upgrade can be performed:

Table 4. Upgrade support matrix

ScaleIO / VxFlex OS / PowerFlex base version	Upgrade path	Supporting documentation
v2.0.1.4, v2.6.x	Two-step upgrade flow: 1. Upgrade to the latest version of v3.0.1.x 2. Upgrade to v3.6.0.x	*Upgrade VxFlex OS to v3.0.x Guide* *Upgrade PowerFlex to v3.6 Guide*
v3.0, v3.0.0.x, v3.0.1.x, v3.5.x, v3.6, v3.6.0.x	Single-step upgrade from: • VxFlex OS v3.0/v3.0.0.x/v3.0.1.x • PowerFlex v3.5.x • PowerFlex v3.6.0.x to PowerFlex v3.6.3	*Upgrade PowerFlex to v3.6 Guide*

CloudLink support

PowerFlex version	CloudLink version	Description of change
3.6.0.6 and later	7.1.9	Existing CloudLink install base with the original device path logic: /dev/sd*X*.
3.6.0.2, 3.6.0.3, 3.6.04, and 3.6.0.5	7.1.3	CloudLink 7.1.3 enables a new device path logic for encrypted devices, using /dev/disk/by-id/wwn-*<unique_id>*.
3.0.1.x all versions 3.5.x.x, all versions 3.6 3.6.0.1	7.1 7.0	The CloudLink versions maintain legacy support for the existing CloudLink install base with the original device path logic: /dev/sd*X*.

PowerFlex 3.6.0.2 supports migration and upgrade flows from CloudLink 7.1 to CloudLink 7.1.2 or CloudLink 7.1.3.

Following are the expected upgrade flow for systems with CloudLink:

- Upgrade PowerFlex to 3.6.0.6.
- For CloudLink 7.1 / 7.0, upgrade to CloudLink 7.1.5.
- For CloudLink 7.1.2 / 7.1.3 / 7.1.4, upgrade to CloudLink 7.1.5 with support assistance.

PowerFlex 3.6.0.6 supports migration and upgrade flows from CloudLink 7.1 to CloudLink 7.1.5.

CloudLink 7.1.5 changes the disk-mapping logic to use /dev/sd*X* compared to what was introduced in the CloudLink 7.1.2 to 7.1.4 versions. This version removes support for /dev/disk/by– id from CloudLink which reverts to a /dev/sd*X* device mapping.

Customers who migrated to CloudLink 7.1.3 and adopted /dev/disk/by-id logic must open a support call as documented in *Dell PowerFlex v3.6.x CloudLink for SDS Devices*. This helps in reverting to /dev/sd*X* device mapping and helps adopt the CloudLink 7.1.5.

Customers using the earlier CloudLink releases (CloudLink 7.1.1 or earlier) can adopt the CloudLink 7.1.5 by using the existing device mapping.

Due to key updates, it is recommended that you adopt the latest CloudLink 7.1.5 for usage with PowerFlex for Data at Rest Encryption.

Supported operating systems

The user documentation contains a list of supported operating systems. For the most current information, see the *Dell Technologies Simple Support Matrices* at https://elabnavigator.dell.com/eln/modernHomeSSM, **Storage** option.

SDC compatibility

By design, the PowerFlex SDC in this release is compatible with previously released systems that are supported (for example, VxFlex OS v3.0.1). In addition, the PowerFlex system (backend) in this release is backward compatible with previously released SDC versions that are supported.

Some limitations may apply to mixed-version systems, such as:

- Replication is not supported on SDC releases earlier than PowerFlex v3.5.
- The CHAP component authentication feature cannot be enabled unless every MDM, SDC, and SDS in the system is at least version 3.5, and conversely, if the feature is enabled, no pre-v3.5 SDCs can be added.
- Fine Granularity Storage Pool-based volumes cannot be mapped to a ScaleIO/VxFlex operating system v2.x SDC.
- Snapshot Policy management is not available with SDC releases earlier than VxFlex operating system v3.0.

Known issues and limitations

The following table lists known issues and limitations that exist in PowerFlex 3.6.3.

NOTE: If an issue was reported by customers, the customers' Service Request numbers appear in the "Issue number & SR number" column, and serve to correlate between customer-reported issues and the PowerFlex issue number.

Table 5. Known issues and limitations—PowerFlex Custom Node

Issue number & SR number	Problem summary	Workaround
SCI-62961	Disk slot information for PowerEdge nodes that are introduced in PowerFlex 3.6 does not work on R650/R6525/R750 nodes with HBA355.	Use iDRAC or perccli to get the disk slot information.

Table 6. Known issues and limitations—PowerFlex Gateway

Issue number & SR number	Problem summary	Workaround
SCI-66203 SR# v989309	The AutoComplete attribute is not disabled for passwords in form-based authentication. If the browser is used in a shared computing environment where more than one person may use the browser, "autocomplete" values may be retrieved or submitted by an unauthorized user.	None
SCI-58953	When running the PowerFlex Gateway on an Internet Explorer 11 browser, LIA Authentication gets stuck in "Loading..." status when retrieving system topology.	Use Chrome or Firefox browsers to perform this action.
SCI-58471	PowerFlex Gateway Tomcat log rotations do not work after an upgrade. At build time, the "server.xml" file is defined as a "noreplace-config file" in the RPM. During an upgrade, the RPM mechanism will automatically detect if the file was previously edited or not. If the file was edited, it will not be overwritten during upgrade, and the new file will be created with the ".rpmnew" suffix.	Edit the file "server.xml" manually, and change the log rotation rules. For more information, see the PowerFlex Troubleshooting Guide, "Some PowerFlex Gateway log files fail to rotate" topic.
SCI-57810	The PowerFlex Gateway Maintenance operation screen might show an invalid LIA authentication for a containerized SDC when connecting to the system. This should appear as N/A, and can be ignored.	None
SCI-57304	PowerFlex Installer (also known as the PowerFlex Gateway) is exposed to issue HSTS Missing From HTTPS Server, CVE: https://nvd.nist.gov/vuln/detail/CVE-2017-7789	None
SCI-54887	When performing the log collection using the PowerFlex Gateway with the "exception only" check box marked, the log collection fails.	Perform log collection with the default settings using the PowerFlex Gateway.
SCI-54799	If Auto Collect logs is configured in the system during an NDU session, when the NDU ends, the PowerFlex Installer displays a button asking to start the first phase of the NDU, even though the previous one was completed successfully (instead of displaying the "Mark operation as done" button).	Disable auto log collection prior to starting the NDU, and run the NDU. After NDU completion, enable auto log collection again.

Table 7. Known issues and limitations—MDM

Issue number & SR number	Problem summary	Workaround
SCI-60885	When using SRA, the PowerFlex Gateway password and the MDM password must be the same. Otherwise. one of the following errors is displayed on the SRM/SRA UI: SRA command 'discoverArrays' failed. General exception No further information available SRA command 'discoverArrays' failed. general error check logs for detail	Change the PowerFlex Gateway and MDM user passwords simultaneously. Ensure that they have admin permissions and the same password.
SCI-69646	By default, Python3 is installed on Ubuntu and is located in `/usr/bin/python3`. During the execution of the command `/opt/emc/scaleio/mdm/diag/dump_and_balance_check.py --username XXXXX --password YYYYY --approve_certificate --scripts_path=/opt/emc/scaleio/mdm/diag/`, the following error is encountered: `bash: /opt/emc/scaleio/mdm/diag/dump_and_balance_check.py: /usr/bin/python: bad interpreter: No such file or directory.`	Add "3" at the start of the Python script.

Table 8. Known issues and limitations—Presentation server

Issue number & SR number	Problem summary	Workaround
SCI-60813	When connecting a presentation server from PowerFlex version 3.6 to a PowerFlex 3.5 system, some of the UI functionality does not work as expected.	Use the matching version of the presentation server. In an upgrade flow, see the upgrade guide for guidelines about when and how to upgrade the presentation server.

Table 9. Known issues and limitations—SDC

Issue number & SR number	Problem summary	Workaround
SCI-57752	The SDC rpm package for SLES15.X installs without any issues. However, after installation, when checking if the SDC driver's scini service is loaded, the following confusing state is returned: service scini status service: no such service scini This is a false positive status alert.	None
SCI-69559	SDC installed with flash vSphere plug-in fails without any recovery due to end of support by VMware for ESXi 6.7.x and 6.5.x. For details, see Announcing end of general support for vSphere 6.5.x and vSphere 6.7.x (89305).	None
SCI-71000	The SDC system service in SLES15.x is renamed to **scini** to maintain consistency with other Linux distributions. However, when upgrading to the latest version (scini.service), the earlier	Ensure to disable and remove the earlier SDC service manually and then, restart the latest scini service. Perform the following procedures:

Table 9. Known issues and limitations—SDC (continued)

Issue number & SR number	Problem summary	Workaround
	version (sdc.service) remains in use. This conflict prevents the SDC from loading.	1. systemctl disable sdc.service 2. rm /usr/lib/systemd/system/sdc.service 3. systemctl restart scini.service

Table 10. Known issues and limitations—SDR

Issue number & SR number	Problem summary	Workaround
SCI-62047	A replication failover on an RCG that is still in the initial copy phase may not present all volume members at the remote target location.	Wait for all volumes in the RCG to complete their initial copy phase before performing the failover operation for that RCG.

Table 11. Known issues and limitations—SDS

Issue number & SR number	Problem summary	Workaround
SCI-63216	When adding NVMe devices using the by-id path similar to the following: dev/disk/by-id/nvme-Dell_Express_Flash_NVMe_P4610_1.6TB_SFF_BTLN950106ZB1P6AGN the MDM blocks the addition due to the error: new_device_path too long, MDM supports up to 63 characters.	Use the alternative name path in by-id location with nvme-eui in the path. For example: /dev/disk/by-id/nvme-eui.01000000010000005cd2e448b0b75151

Table 12. Known issues and limitations—vVols

Issue number & SR number	Problem summary	Workaround
SCI-54057	Unmapping a VMware virtual volume, and then immediately afterwards trying to overwrite its content, might generate the error: "volume n is in use".	None. Try again shortly after, because some volume operations are performed in an asynchronous fashion.

Operating system known issues and limitations

Learn about known issues and limitations for specific operating systems.

Table 13. Known issues and limitations—operating systems

Issue number & SR number	Problem summary	Workaround
VXFV-296	Mixing the Mellanox CX-6 100GbE NIC driver with the Broadcom PCIe/OCP 25GbE NIC driver is not supported.	Remove the `bnxt_en` driver and install the Mellanox driver.
VXFV-283	iDRAC support for "Prepare for Removal" on PowerEdge R650/R6525/R750/ nodes fails.	NVMe hot-swap on Linux operating systems is supported. Skip the Prepare for Removal step in iDRAC.
VXFV-265	During execution of the command `create-namespace` with 4k alignment on RHEL 8.4, the following error is encountered: Warning: namespace0.0 unable to validate alignment	This is a false notification. The command result is as expected, and the resulting DAX device can be used with PowerFlex acceleration devices.

Table 13. Known issues and limitations—operating systems (continued)

Issue number & SR number	Problem summary	Workaround
VXFV-360	The device name appears incorrectly as Non PCI device instead of iDRAC, but the current and baseline versions are accurate (6.00.30.201 and 6.00.30.202, respectively).	Dell recommends upgrading the OME catalog to the latest version.
—	PowerFlex nodes running CPLD 1.0.7 stop responding during Power On Self-Test (POST) when upgrading to certain RCM or IC packages that include BIOS 1.8.2 and iDRAC 6.00.30.202.	Before upgrading to the latest supported version of PowerFlex, ensure that the PowerFlex nodes are running CPLD 1.0.8 or later. For details, see https://www.dell.com/support/kbdoc/en-us/000205925.

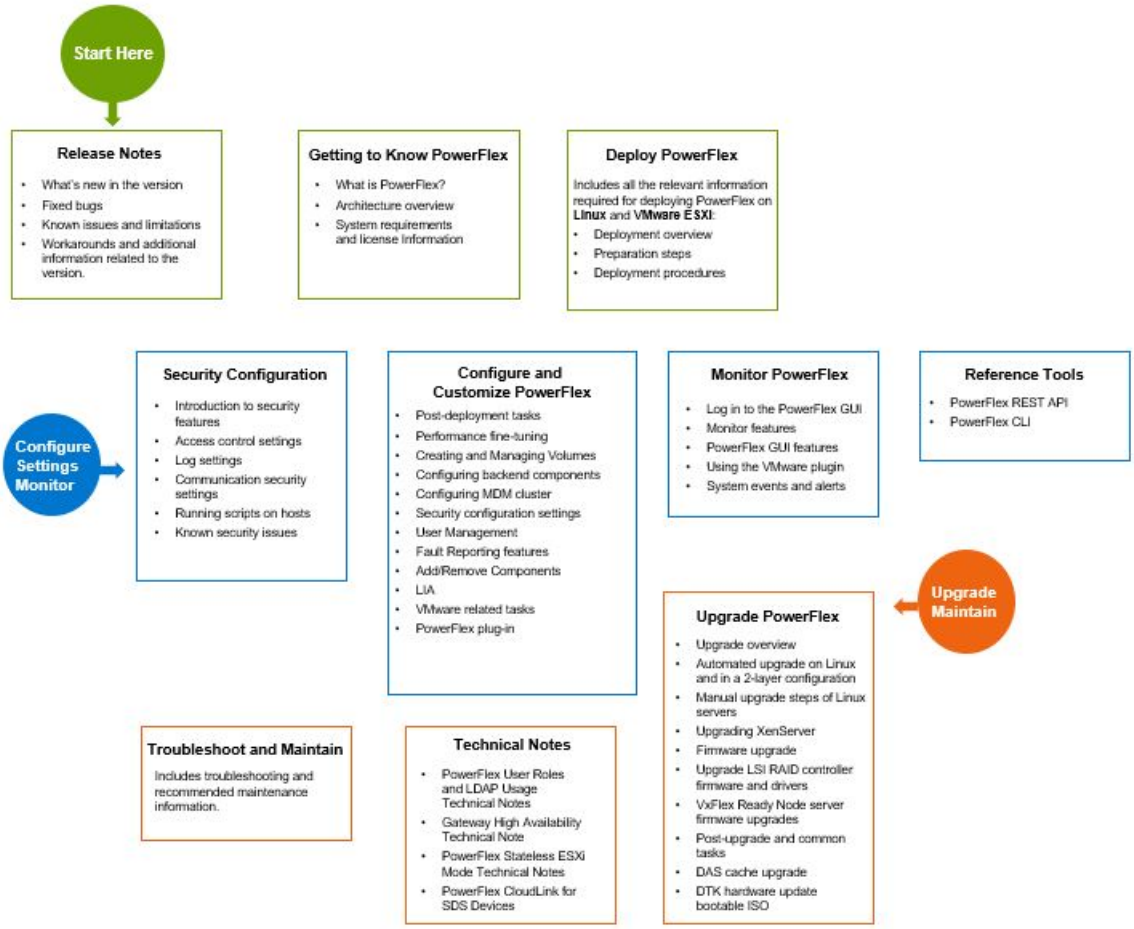
Additional resources

Use these resources to find more information about this product, get support and provide feedback.

Product information

For documentation, release notes, software updates, or information about Dell products, go to Dell Online Support at:

https://www.dell.com/support/home/en-us/product-support/product/scaleio/overview



Where to get support

Go to Dell Online Support and click **Support**. You will see several options for contacting Dell Technical Support. Note that to open a service request, you must have a valid support agreement. Contact your Dell sales representative for details about obtaining a valid support agreement or with questions about your account.

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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